Gears and wheels

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use gear ratio and wheel circumference to predict an ideal travel speed.

SUCCESS CRITERIA - I can...

1. Connect gear teeth with speed ratio.
2. Calculate distance per wheel revolution.
3. Explain the tradeoff between ideal speed and available torque.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “gear ratio” and “circumference” to explain your model or result. Sentence starters are optional.

gear ratio: a comparison of gear tooth counts or rotation rates

circumference: distance around the edge of a circle

torque: the turning effect of a force about an axis

01 Warm-up: bring back an earlier idea

A 30-turn coil has flux per turn change by 0.005 Wb in 0.3 s.
Find average induced-emf magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

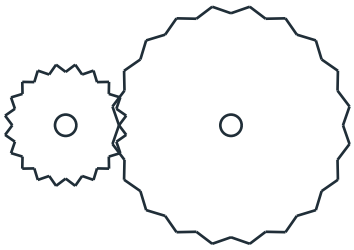
02 Worked example: explain the model

Reference: $rpm_{out} = rpm_{in} \times teeth_{driver}/teeth_{driven}$; $C = \pi d$
$v = C \times rpm_{wheel}/60$.
Ideal gears and rolling without slip; C in meters gives v in m/s. Loading and losses change real performance.

Worked reasoning: A 10-tooth driver turns a 30-tooth driven gear. The driven gear makes one turn for every three driver turns. In an ideal gear pair, slower rotation trades for greater torque.

Your turn: Draw both gears, label tooth counts and opposite rotation directions.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Driver Driven

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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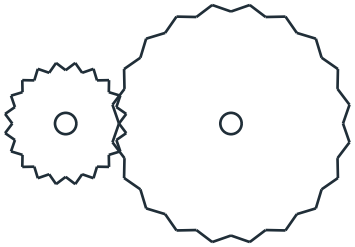
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A small driver gear turns a larger driven gear.
Predict which turns more slowly and indicate their opposite rotation directions.

Sentence starter: Both … . They differ in … because … .



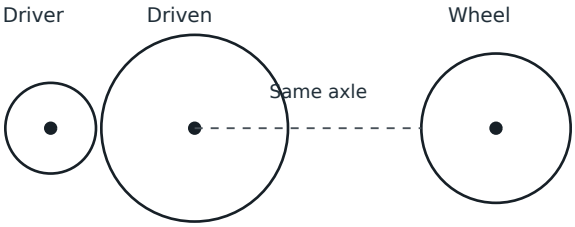
Marzano check - circle: 1 2 3

04 Apply the physics

A model-car motor turns at 600 rpm. A 10-tooth driver turns a 30-tooth driven gear attached to a wheel of diameter 0.060 m.
Label teeth, wheel size and rotation directions.
Calculate wheel rpm and ideal ground speed with no slip.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.



Label teeth, rotation rates and wheel size.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A team chooses the largest wheels because the formula predicts more distance per turn.
Explain why measured acceleration or speed might still decrease.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 7 Ω resistor is in series with two parallel 10 Ω resistors across 24 V.
Find parallel equivalent, total resistance and supply current.
Draw the stated circuit and label quantities relevant to the question.

Word help: series = connected along one unbranched current path

Sentence starter: I found … [unit]. This value means … in this situation.

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

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Continue down each column.

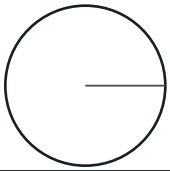
Sentence starters are optional. You may plan in any language.

07 Exit check

A 12-tooth driver turns a 36-tooth gear.
Find ideal output torque as a multiple of input torque and output speed as a fraction of input speed.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.



Radius / diameter: ____

Turns per second: ____

Travel per turn: ____ Ground speed: ____

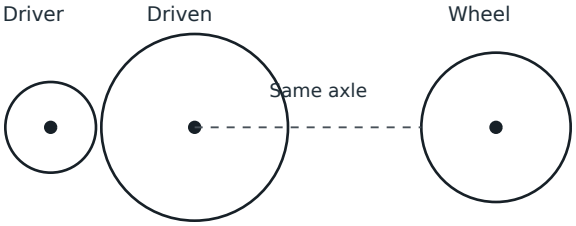
Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 12-tooth driver at 900 rpm turns a 36-tooth gear attached to wheels of radius 0.03 m.
Find ideal wheel rpm and ground speed without slip.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.



Label teeth, rotation rates and wheel size.

Marzano check - circle: 1 2 3